CCC-C-442b
May 12, 1965

SUPERSEDING
Fed. Spec. CCC-C-442a
April 20, 1961

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON, BLEACHED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bleached cotton duck utilized in the manufacture of clothing items (see 6.1).

1.2 Classification.

1.2.1 *Types.* The bleached cotton duck shall be of the following types as specified (see 6.1 and 6.2):

Type I—5 ounces per square yard.
Type II—7 ounces per square yard.
Type III—8.2 ounces per square yard.
Type IV—8.5 ounces per square yard.
Type V—7.0 ounces per square yard.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-T-191—Textile Test Methods.
PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

FSC 8305

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(Application for copies should be addressed to the Federal Trade Commission, Washington, D. C., 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is required, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 *Yarn.* The yarn shall be made from cotton which has been cleaned, carded, drawn, spun, and twisted into singles or plied yarns as specified in table I.

3.3 Color. The color shall be chemically white. The chemically bleached white may be supplemented with fluorescent optical brightener. The optically brighter finished cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.4. When no standard is available, the discoloration shall be not less than "fair" when tested as specified in 4.4.

3.3.1 *Matching.* The color shall match the available standard sample (see 6.3) under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

3.5 Weave. The weave for types I, III, and V cloths shall be plain, one up, one down. The weave for types II and IV cloths shall be an oxford weave with two ends weaving as one and alternating in plain-weave order with each pick.

3.6 Finish. The cloth shall be desized, singed, bleached, and mercerized.

3.6.1 *Nonfibrous materials.* The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.4.

3.7 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 2.0 percent in either the direction of the warp or filling when tested as specified in 4.4. The preshrinking process used shall not be identified in any manner on the cloth, the ticket, or on the package.

3.8 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width, inclusive of selvages.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous lengths, each not less than 50 yards. The cloth shall be put up on rolls as specified in PPP-P-51.

3.10 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The occur-

TABLE I—*Physical requirements*

Type	Weight per square yard minimum	Yarn ply		Yarns per inch minimum		Breaking strength minimum	
		Warp	Filling	Warp	Filling	Warp	Filling
	Ounces					*Pounds*	*Pounds*
I	5.0	2	2	77	44	130	75
II	7.0	1	2	88	28	110	60
III	8.2	2	2	58	48	135	100
IV	8.5	1	4	112	32	120	120
V	7.0	2	2	52	34	80	48

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rence of defects shall not exceed the applicable point level specified.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.2.1 *Testing of components.* Quality assurance provisions for components described shall be in accordance with subsidiary specifications referenced to the extent applicable. The yarn requirement as specified in 3.2.1 will be accepted on the basis of a supplier's certificate of compliance.

4.2.2 *Yard-by-yard examination.* The required yardage of each roll shall be examined on the face side and all defects as defined in section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1, except only those slubs and knots which exceed the maximum acceptable limits shown for each in figure 1 shall be scored. None of the 30 contiguous yards (increments of 1 yard on the measuring device) selected for examination from any one roll within the sample shall be penalized more than 4 points. The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 rolls. An approximately equal number of yards shall be examined in each roll of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 35.00. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of material (inches)}} = \text{Points per 100 square yards}$$

4.2.2.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to and including 3 inches in any dimension	one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	three points
For defects exceeding 9 inches in any dimension	four points

Note: The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Width less than specified.
Overall uncleanness.
Objectionable odor.

4.2.3 *Examination for length.*

4.2.3.1 *Individual rolls.* During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.3.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.3.3 *Examination for identification of shrinkage process and compliance with Tex-*

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tile Fiber Products Identification Act. During the yard-by-yard examination each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample contain identification of the shrinkage processes by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-51 to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with.

4.4 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards, full width of the finished cloth. The sample size shall be as listed in table II. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units fail to meet any requirements specified. When the data in the "number of determinations per sample unit" and the "results reported as" columns are not specified in table III they shall be as required by the referenced test method.

TABLE II

Lot size (units)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A and C.* The cloth, put up as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

TABLE III—*Test methods*

Characteristics	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as
Weight, ounce/square yard	3.4	5041	—	—
Yarns per inch	3.4	5050	—	—
Breaking strength	3.4	5100	—	—
Yarn ply	3.4	Visual	1	Pass or fail
Discoloration	3.3	[1]5660	—	—
Weave	3.5	Visual	1	Pass or fail
Mercerization	3.6	Microscopic	1	Pass or fail
Nonfibrous material	3.6.1	2611	—	—
Shrinkage in laundering	3.7	5550	—	—

[1] Method 5660 shall be used, except that exposure of the specimen and standard sample shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "satisfactory." Any specimen showing discoloration greater than that of the standard sample shall be rated "unsatisfactory." When no standard sample is available the discoloration of the specimen shall be rated "fair," when there is an appreciable but not objectionable change in color. One determination shall be made on each sample unit and the result reported as "pass" or "fail."

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5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP-P-51.

5.3 **Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-51.

6. NOTES

6.1 **Intended use.** Types I, II, and III bleached cotton ducks covered by this specification are intended for use in white uniforms for personnel of all Government departments. Type IV bleached cotton duck is intended for use in dress uniform trousers of U.S. Military Academy Cadets. Type V bleached cotton duck is utilized as the base cloth for Smock, Man's, Dental, Operating; and Smock, Man's, Medical Assistant.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and nomenclature of cloth required (see 1.2).
(*c*) Width required (see 3.8).
(*d*) Length of pieces if other than specified (see 3.9).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives as indicated in PPP-P-51 may be exercised in the preparation for delivery of cloth.

6.3 **Standard sample.** For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Level B packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

6.5 **Transportation description.** The transportation description and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods, finished, not otherwise indicated by name.
Carload minimum weight—Any quantity.

Motor:

Cotton piece goods, not otherwise indicated.
Truckload minimum weight—Any quantity.

CUSTODIANS:
Army—GL
Navy—SA
Air Force—11

Review activities:
Army—MD
Navy—SA, MS
Air Force—11, 67

User activity:
Navy—MC

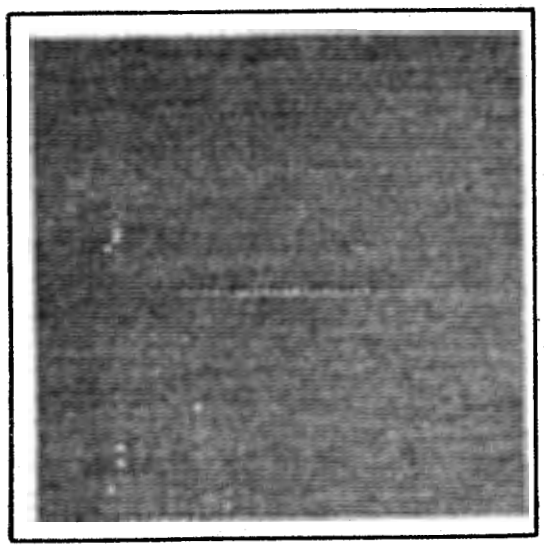
Preparing activity:
Army—GL

Review and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing Of DOD Standardization Documents.

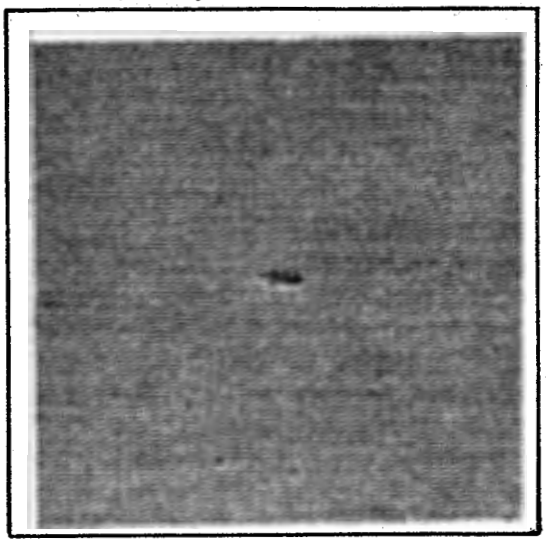
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FIGURE 1. Limits of acceptability

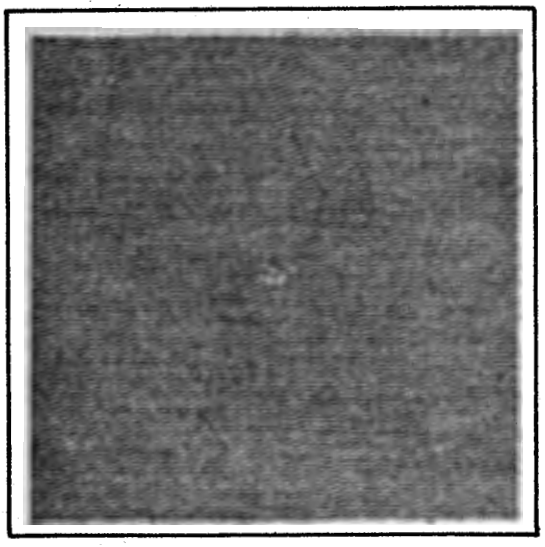
Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)



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